

International Journal of Sustainable Transportation

ISSN: 1556-8318 (Print) 1556-8334 (Online) Journal homepage: www.tandfonline.com/journals/ujst20

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To cite this article: Vijay Raviprabhakaran, Jesica Sarakonda, Vinay Udugula & Nishitha Varma Datla (21 Nov 2025): Electric automobile route scheduling with time setting using machine learning methods, International Journal of Sustainable Transportation, DOI: 10.1080/15568318.2025.2586217

To link to this article: https://doi.org/10.1080/15568318.2025.2586217

Published online: 21 Nov 2025.

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INTERNATIONAL JOURNAL OF SUSTAINABLE TRANSPORTATION
https://doi.org/10.1080/15568318.2025.2586217

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Electric automobile route scheduling with time setting using machine learning methods

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ABSTRACT

Electric automobiles transform urban mobility by offering a clean and eco-friendly alternative to traditional internal combustion vehicles. Planning and optimizing electric automobile (EA) travel routes involve considering energy consumption, battery range, and the availability of charging stations. By optimizing routes that account for delivery deadlines and charging outlet availability, the Electric Automobile Path Planning with Time Frame (EAPPTF) aims to reduce travel time and energy consumption. While decision tree models forecast and optimize energy consumption and charging stops, route planning uses Dijkstra's algorithm to identify the shortest path. Using real-time data from Indian cities, this article enhances electric automobile routing by providing insights into traffic flow and charging station availability, thereby enabling efficient travel and energy management. Data from Bengaluru and Hyderabad demonstrate that the proposed methods effectively address the routing challenges for electric automobiles, resulting in shorter trips and lower energy consumption, thereby promoting sustainable urban mobility. By improving autonomous driving technologies and charging infrastructure, these algorithms can also boost urban mobility and sustainability. This strategy supports optimal vehicle operation while addressing pressing issues in emerging nations.

ARTICLE HISTORY
Received 7 January 2025
Revised 13 August 2025
Accepted 31 October 2025

KEYWORDS
Electric vehicle route transportation; time-constrained route scheduling; machine learning for EV routing; Dijkstra algorithm; optimal charging station placement

1. Introduction

Electric automobiles (EAs) are gaining popularity in emerging countries, especially in urban transportation and logistics, because they can reduce emissions and promote sustainability. As a result, the Electric Automobile Routing Problem (EARP) has been extensively studied, with a focus on energy efficiency, battery management, and route optimization. EARP is crucial for developing eco-friendly mobility solutions. Although it does not directly address practical issues, the Vehicle Routing Problem with Time Windows (VRPTW) utilizes reinforcement learning to optimize specific constraints (Jia et al., 2025; Lin et al., 2018). A hybrid ALNS/TS heuristic has been applied to enhance electric vehicle routing with time windows, thereby improving delivery efficiency and sustainability (Liang et al., 2024). Another model considers fluctuating electricity prices and proposes two cost-saving strategies: charging during off-peak hours and returning power to the grid during peak hours. Numerical experiments with real-world data analyze how energy costs, service hours, and other factors affect Electric Vehicle (EV) fleet operations (Coppo et al., 2025). The importance of data distribution in training models is highlighted in Improving Combinatorial Optimization with Machine Learning, which advocates for combining machine learning with optimization techniques (Li, 2024). Reinforcement learning balances centralized decision-making and decentralized learning to enhance the operation of community-owned EA fleets, reducing costs, electricity bills, and customer wait times (Si et al., 2024). An attention-based adaptive critic integrated with deep reinforcement learning (DRL) handles automobile routing problems more effectively by combining DRL with local search, improving solution quality and performance (Lin et al., 2025). The Exploratory Combinatorial Optimization-Deep Q-Network (ECO-DQN) presents a novel reinforcement learning approach for graph combinatorial problems, demonstrating improved performance when combined with other search strategies (Daliri et al., 2024). Reinforcement learning (RL), through adaptive cutting plane selection, can also boost the performance of integer programming solvers, making the Branch-and-Cut algorithm more efficient and outperforming traditional heuristics across various problems (Hu et al., 2025). Even without problem-specific training, Neuromata, a message-passing neural network, is trained as a satisfiability (SAT) classifier and adapts well to complex SAT tasks like clique recognition and graph coloring (Chang et al., 2024). Graph neural networks (GNNs) are used to solve the traveling salesperson problem (TSP), providing robust but approximate solutions by estimating costs within a near-optimal range (Xu et al., 2025). When combined with effective Q-learning and a probabilistic greedy mechanism, Graph Convolutional Networks (GCNs) efficiently solve graph-based combinatorial problems, offering high-quality and cost-effective alternatives to advanced algorithms (Daliri et al., 2024). A model

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that integrates attention mechanisms with reinforcement learning and greedy rollouts improves learned heuristics for problems like TSP, Vehicle Routing Problems (VRP), Orienteering Problems (OP), and Prize Collecting Traveling Salesman Problems (PCTSP), producing solutions close to optimal (Li et al., 2024). For autonomous buses in dedicated lanes, a data-driven cooperative adaptive cruise control (CACC) method based on reinforcement learning helps maintain consistent journey times based on bus volume data (Liang et al., 2025). To tackle NP-hard problems, a scalable graph convolutional network predicts vertex likelihoods to find optimal solutions, outperforming current deep learning methods and matching top heuristics (Cui et al., 2025). A new reinforcement learning approach for the Vehicle Routing Problem (VRP) can handle different scenarios more effectively and provide higher-quality solutions than existing methods by training a single policy model. This approach may also address other combinatorial optimization challenges (Guezouli et al., 2025). A neural network using policy gradients and reinforcement learning can produce near-optimal solutions for the TSP (Li et al., 2025). A machine learning method for variable branching in mixed-integer programming reduces search tree size and rivals top commercial solvers by learning from strong branching decisions with a surrogate function based on a learning-to-rank strategy (Du et al., 2025). Using branch-price-and-cut algorithms, four variants of the electric automobile-routing problem (EARP) are analyzed to help customers and reduce costs (Miao et al., 2025). A scalable approach incorporates structured data to embed latent variables into feature spaces, creating smaller, faster models with state-of-the-art predictive performance (El Hadraoui et al., 2024).

A neural network called Pointer Networks uses attention as a pointer for output sequences of varying lengths. Convex hulls are incorporated into this network, and the Traveling Salesman problem is a geometric challenge that is better addressed than with existing methods (Chung et al., 2025). Instead of relying on a fixed-length vector, neural machine translation allows the model to identify relevant passages in the original text. When using this approach, alignments correspond with social perception, and translation performance is comparable to leading novel expression-based algorithms (Zhang et al., 2025). A systematic approach to integrating power system simulations and PEV electricity consumption demonstrates the effects of EAs and plug-in hybrid electric vehicles on the electrical network. Focusing on efficient last-mile delivery route planning, it addresses the problem of the EA route (Yu & Anh, 2025). Deep neural network training employs gradient descent and random initialization, emphasizing how activation functions influence nonlinear behavior. It proposes a novel initialization strategy that improves convergence by overcoming saturation issues and maximizing the Jacobian singular values during training (Lv et al., 2025). The EA Routing Problem with Time Windows (E-ARPTW) and recharging stations are addressed through a combination of tabu search and variable neighborhood search. It evaluates how charging strategies impact electrical system bottlenecks. Experimental results support the usefulness of heuristics in practical logistics (Pareek et al., 2024). An evaluation model for shared vehicle systems emphasizes vehicle availability, distribution, and energy management. Waiting times for an optimal number of vehicles are reduced in the Southern California case study (Fan et al., 2025). Workshop scheduling is integrated with reinforcement learning, and a heuristic evaluation function is optimized using a repair-based scheduler with Temporal Difference (TD) learning (Raviprabhakaran et al., 2024). Due to the problem's complexity, approximation methods are explored for scheduling and routing vehicles with time window constraints. Extensive computer studies show that several heuristics perform well, with an insertion-type heuristic consistently producing excellent results across various scenarios (Chang et al., 2025). Optimization methods previously used to solve the routing problem often get stuck in local optima and thus do not always find a global solution (Raviprabhakaran, 2022, 2023a, 2023b; Vijay, 2018; Vijay & Subramanian, 2016). To tackle optimization challenges like the Traveling Salesman Problem, massively interconnected networks employ nonlinear analog neurons. These networks demonstrate that they can efficiently solve complex combinatorial problems by processing responses with analog inputs quickly (Raviprabhakaran, 2024). Several approaches can be explored for the EA Routing Problem based on existing research; however, no real-time data analysis has been conducted yet.

This study examines real-time scenarios in the two most populous cities and provides recommendations for optimal consumer locations and routes. The decision tree approach is introduced for Electric Automobile Path Planning with a Time Frame, providing a structured and understandable framework for decision-making in complex routing situations. It breaks down the problem into hierarchical decisions based on key variables, such as vehicle battery levels, charging station availability, customer locations, and time-window limits. Decision trees also incorporate dynamic constraints such as real-time traffic and charging station occupancy to adjust routes. Their flexibility makes them suitable for EA operations, allowing for the integration of vehicle-specific limits, fluctuations in electricity prices, and energy consumption patterns. To improve performance, scalability, accuracy, and route efficiency in large-scale EAPPTF scenarios, decision trees are combined with the Dijkstra algorithm and compared with the Random Forest method. The real-time scenarios of Hyderabad and Bengaluru, two Indian cities, are examined. These methods aim to optimize objectives such as lowering operating costs, increasing energy efficiency, and ensuring on-time delivery. This provides a robust solution for modern electric logistics networks in the selected cities.

2. Electric automobile routing problem formulation

To address the operational challenges of EAs, EARP is an improved version of the Automobile Path Planning Problem (APP). While considering battery constraints and energy consumption, the goal is to optimize routes and minimize

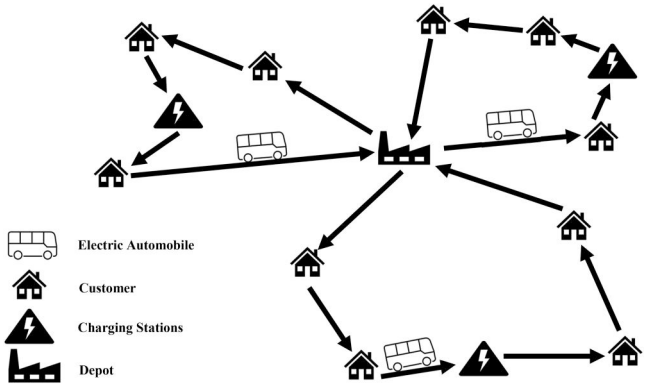


Figure 1. Illustrative example of the electric automobile routing problem.

costs, including travel distance, journey time, charging expenses, and total time. The model also accounts for complexities such as service windows and charging station availability. The illustrative example and operation of the electric automobile routing problem are shown in Figure 1.

EARP supports sustainable logistics in waste collection, public transit, and urban deliveries. It emphasizes key operational factors through restrictions such as battery energy levels, system time updates, and the availability of EAs to meet consumer demands. Additionally, a decision tree model incorporating traffic, EA weight, and other dynamic factors is used to enhance route efficiency.

Primarily, system time stm^{t+1} is rationalized as trials

$$(btt^t)_{req} stm^{t+1} = \begin{cases} Maximum(stm^t, e_{ap^t}) + k + j(ap^t, ap^{t+1}), \ if \ ap^t \in V_{cj} \\ stm^t + (btt^t)_{req} + j(ap^t, ap^{t+1}), \ \ if \ \ ap^t \in V_{cs} \\ j(ap^t, ap^{t+1}), \ \ if \ \ ap^t \in V_{dp} \end{cases} \tag{1}$$

where stm^t is time, k is a persistent signifying the facility time at every customer's apex, $j(ap^t, ap^{t+1})$ is the journey time from one apex ap^t to the next apex ap^{t+1}, $(btt^t)_{req}$ is the amount of energy required to fully charge the battery btt^t, V_{cj} is the customer juncture, V_{cs} is a charging outlet point and V_{dp} is a depot point.

The energy level of the active EA's battery is then apprized as

$$btt^{t+1} = \begin{cases} btt^t - c(ap^t, ap^{t+1}), \ \ if \ \ ap^t \in V_{cj} \\ MBtt - c(ap^t, ap^{t+1}), \ \ otherwise \end{cases} \tag{2}$$

where $c(ap^t, ap^{t+1})$ is the EA energy consumption by moving from one apex ap^t to the subsequent apex ap^{t+1}, $MBtt$ is the maximum battery competence, btt^{t+1} is the apprized live EA's battery level.

Decisively, the number of EAs existing ea^t, and the essential demand dem^t at each apex, are apprized as trails.

$$ea^{t+1} = \begin{cases} ea^t - 1, \ \ if \ \ ap^t \in V_{dp} \\ ea^t, \ \ otherwise \end{cases} \tag{3}$$

$$dem^{t+1} = \begin{cases} 0, \ \ if \ \ q^t = m \\ dem^t, \ \ otherwise \end{cases} \tag{4}$$

ea^{t+1} is updating the quantity of EAs that are available and dem^{t+1} is updating the necessary demand.

$$At = \sum_{(x,y) \in Way} \hat{A}t_{xy} \tag{5}$$

where $\hat{A}t_{xy}$ is the projected amount of time required to go between intersections x and y using the Decision Tree mode and At is the total amount of time spent traveling between the beginning and terminating junctions.

Due to their specific limitations, such as low battery capacity and dependence on charging infrastructure, EAs require careful planning. While client time windows ensure on-time service, energy consumption and the availability of charging outlets determine the operational range and recharging needs. Considering these factors not only makes routes more practical but also improves overall efficiency, allowing vehicles to meet delivery or service commitments without exceeding their operational limits.

2.1. EA routing constrictions

2.1.1. Solitary route accomplishment

$$\sum_{y \in J} R_{xy} = 1, \ \ \forall x \neq J \tag{6}$$

Every junction needs to be visited precisely once, where R_{xy} is $R_{xy} = 1$ if the route departs from the junction x to junction y; otherwise, $R_{xy} = 0$, R_{xy} is the path between x and y.

2.1.2. Path endurance

$$\sum_{y \in J} R_{xy} - \sum_{y \in J} R_{xy} = 0, \ \ \forall x \neq s, J \tag{7}$$

This guarantees that the route remains uninterrupted from the beginning node to the final junction J.

2.2. Time window constrictions

2.2.1. Arrival time constrictions:

Let Atm_x be the advent time at the junction. x within a precise time window $[\psi_x, \zeta_x]$.

$$\psi_x \leq Atm_x \leq \zeta_x \tag{8}$$

where ψ_x is the least time limit, ζ_x is the extreme time limit. The advent time Atm_y at a subsequent junction y.

This is delineated as $Atm_y = Atm_x + \hat{A}t_{xy}$.

2.3. Energy constrictions

2.3.1. Battery competency

Let w_x is the battery's energy level at the junction. x, and l is the battery competent.

$$0 \leq w_x \leq l \tag{9}$$

2.3.2. Edge energy use

W_{mn} is the use of energy to get from the junction x to y, w_y is the energy at y junction, and w_x is the energy at x junction.

$$w_y = w_x - W_{xy} \tag{10}$$

where W_{xy} is estimated or calculated using vehicle and distance characteristics.

2.3.3. Charging outlet accessibility

If a junction x is a charging place, the battery energy w_x increases up to l

$$co_y = Minimum(l, w_m + c_r.Atm_c) \tag{11}$$

where c_r is the charging rate and Atm_c is the charging time, $\mathrm{Min}(l, w_x + C_r.Atm_c)$ is the nearest charging outlet.

3. Dijkstra algorithm and decision tree procedures

In a grid that shows road networks, the fastest route between two points is found using Dijkstra's algorithm. It works by choosing the node closest to the starting point and updating the distances to its neighbors iteratively. Using a greedy approach to expand the shortest path first, this algorithm ensures that the quickest path to a node remains the same once it has been identified. The process continues until the shortest path to all nodes is found. It performs well with grids that have positive edge weights and is often used to find the most efficient routes in applications such as GPS navigation. The decision tree method is a type of supervised learning used for tasks such as regression and classification. It builds a tree structure where each node represents a decision that splits the data into subsets based on selected features. The final output or prediction is given by the tree's last node. This method relies on clear, structured data and is easy to understand. It is useful for decision-making involving many criteria because it provides a transparent view of the process. As shown in Figure 2, decision trees are commonly used in predictive analytics and decision support systems to help make choices based on multiple input factors.

4. Implementation of Dijkstra and decision tree methods in real-time scenarios

In Indian cities like Bengaluru and Hyderabad, where urban issues such as traffic congestion, public transportation, and resource management are prevalent, the Dijkstra algorithm and decision tree techniques are employed to address real-time data challenges. Dijkstra's algorithm is ideal for finding the fastest route between nodes in a network. When combined with real-time data, it dynamically identifies the optimal routes for delivery services by considering traffic density. The entire process of applying Dijkstra's algorithm to the problem under consideration is shown in Figure 3. Decision trees can make decisions based on various input features. They categorize data based on time and traffic patterns to predict traffic levels and suggest alternative routes or modes of transportation. These techniques demonstrate how cities like Hyderabad and Bengaluru use real-time data for smarter urban planning and more efficient daily operations. In grid-based systems, such as Bengaluru's complex road network, the Dijkstra method finds the least-cost route.

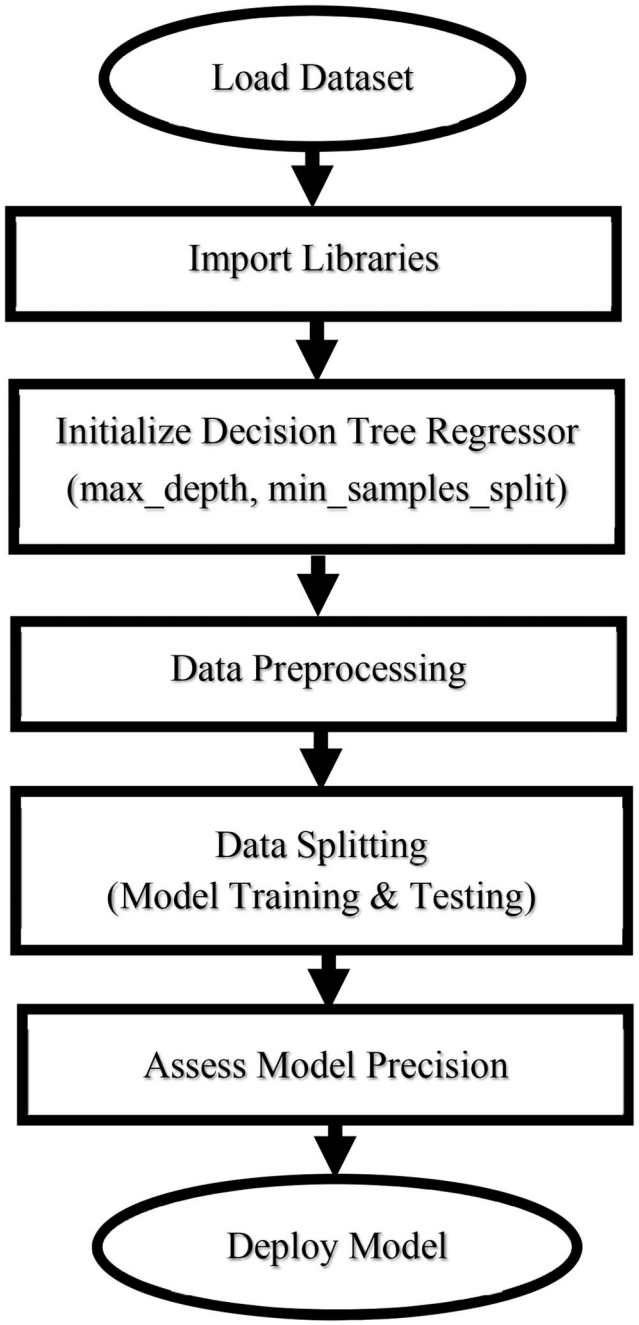


Figure 2. Decision tree regression flow model.

Using GPS and live traffic information, it automatically determines the quickest routes for delivery and emergency vehicles, considering factors such as traffic density, roadblocks, and peak hours. Decision trees, which excel at predictive analysis based on historical data, are used to classify and forecast traffic conditions.

The Dijkstra algorithm and decision trees offer valuable methods for leveraging real-time data in Hyderabad to solve urban challenges, including city planning, traffic congestion, and resource allocation. The Dijkstra algorithm, widely used for identifying the shortest paths in a network, helps optimize Hyderabad's road infrastructure. Incorporating real-time traffic information and considering variables such as construction zones, road closures, and traffic density, it

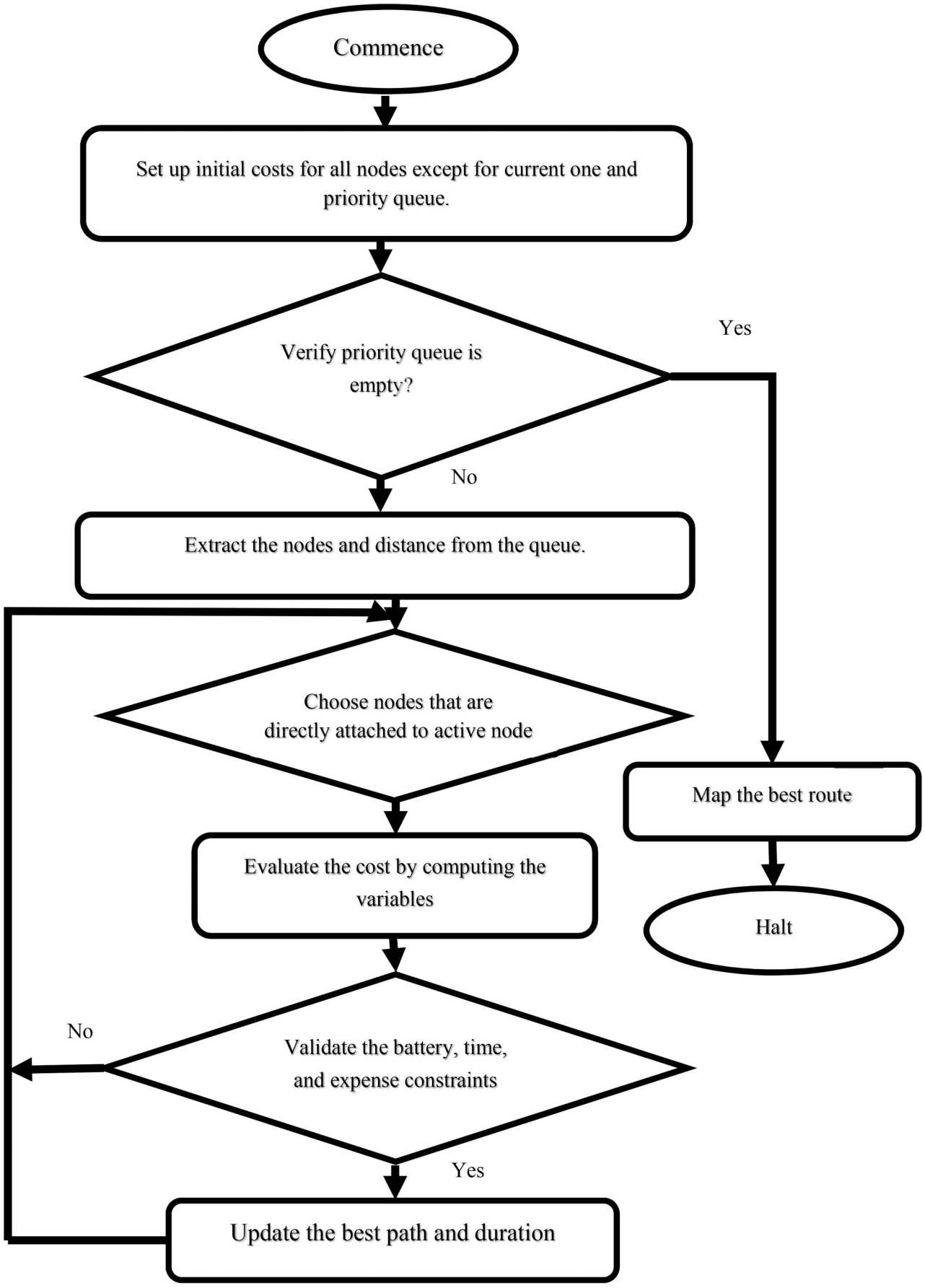


Figure 3. Dijkstra algorithm flow diagram for the EAPPTF problem.

Table 1. Comparison of electric automobile path planning with time frame and decision tree methods.

Aspect	Decision tree regression	Time-frame-based electrical automobile path planning (EAPPTF)
Root node	Factors such as time and distance are often the main attributes used to split the data.	It begins with a starting point, usually the EAs current location or a base station.
Subtrees	Decision procedures generate subsets of input features from the primary factors, further dividing the data into distinct groups.	Decisions or routes diverge based on constraints such as availability, distance, or time limits.
Splitting criteria	This approach relies on reducing entropy, the Gini index, or minimizing variance (in regression).	It makes divisions based on meeting time constraints, optimizing charging schedules, or reducing travel time.
Leaf nodes	The outcome includes final expected values, such as travel costs or charging demand.	The optimal charging outlet and path selection is the result of path planning.
Feature selection	Important features including time, demand, and distance are considered when selecting variables.	Factors including station proximity, time frames, and traffic patterns help evaluate decisions.
Tree depth	The process is influenced by the complexity of the data and the criteria applied for separation.	This depends on the number of decision steps, including locations, charging outlets, and time windows.
Decision outcome	The intricacy of the data and the rationale for its division determines the procedure.	It is dynamic and real-time, adjusting to changes in EA location, traffic, or station availability.
Adaptability	The model remains unchanged until new data is used to update the training.	It adapts in real-time to shifts in station availability, traffic, or EA location.

enables dynamic route planning for delivery services. Decision trees assist in predicting traffic patterns, recommending alternative routes, and improving public transportation efficiency by forecasting and analyzing outcomes based on multiple inputs.

5. Execution of decision trees and the Dijkstra algorithm for time-based electric automobile path planning

5.1. Relating decision tree to time-based electric automobile path planning

This table illustrates the relationship between EA path planning and decision tree regression when a time frame is applied to a specific aspect. Using inputs such as distance, traffic, battery levels, and customer time windows, decision tree regression can be employed to predict or optimize certain variables in the path planning process, such as trip duration, charging schedules, or delivery deadlines. A detailed examination of how these predictions relate to the actual application of time-sensitive EA routing is presented in Table 1.

This Table 1 highlights the connection between decision tree regression and its role in EA Path Planning, particularly concerning time constraints across various operational aspects. Decision tree regression is a technique used to forecast and enhance key elements, including expected trip times, energy recharging scheduling, and deadline compliance, based on input parameters like route distances, traffic conditions, battery status, and customer time windows.

By applying this method, routing decisions can be thoroughly analyzed to improve efficiency and reliability. Comparing the predictions with real operational needs underscores the model's importance in time-sensitive applications. Exploring these relationships demonstrates how effectively the method addresses challenges in EA-based logistics. These insights support efforts to improve system performance and routing accuracy.

The primary objective of the complex optimization problem known as EAPPTF is to identify the most efficient routes for EAs to deliver products or services. This involves considering factors like battery capacity, charging needs, and specific time windows for each customer. An approach combining Dijkstra's algorithm with decision tree regression is used to tackle this problem. We utilize Dijkstra's method to find the shortest paths between nodes, accounting for time windows and battery constraints. To optimize routes and reduce charging stops, decision tree regression predicts energy consumption based on variables such as speed, road elevation, and traffic conditions. By integrating these approaches, our work provides reliable and effective solutions for EA route planning.

5.2. Dijkstra's algorithm for time-frame-based electric automobile way scheduling

5.2.1. Graphical demonstration

Displaying road maps as a weighted graph with edges representing variables like journey distance (D) and apexes denoting locations (such as delivery sites or charging outlets), time (Atm), or energy consumption (W_{xy}).

5.2.2. Integrate constrictions

Adapt edge weights to battery restrictions (Btt), charging standards ($(btt)_{req}$), and time window boundaries $[\psi_x, \zeta_x]$, making sure routes satisfy operational requirements.

5.2.3. Shortest viable conduits

Determine the shortest routes to cut down on journey time (Tm) and energy usage W_{xy} while considering possible recharging pauses.

Table 2. Main charging outlets and client sites in Bengaluru city requiring EA services.

Charging outlet points	Charging outlet locations	Customer points	Customer locations
SP 1	13.0033, 77.5779	**CP 1**	12.9988, 77.5730
SP 2	12.9877, 77.5761	**CP 2**	12.9855, 77.5724
SP 3	12.9880, 77.5880	**CP 3**	12.9840, 77.5915
SP 4	12.9767, 77.5880	**CP 4**	12.9745, 77.5909
SP 5	12.9773, 77.5774	**CP 5**	12.9743, 77.5755
SP 6	12.9833, 77.6036	**CP 6**	12.9828, 77.6017
SP 7	12.9573, 77.5657	**CP 7**	12.9590, 77.5631
SP 8	12.9496, 77.5729	**CP 8**	12.9520, 77.5730
SP 9	12.9576, 77.5837	**CP 9**	12.9593, 77.5864
SP 10	12.9660, 77.5931	**CP 10**	12.9634, 77.5963
SP 11	12.9681, 77.5944	**CP 11**	12.9709, 77.5993
SP 12	12.9735, 77.5969	**CP 12**	12.9731, 77.6111
SP 13	12.9780, 77.6023	**CP 13**	12.9794, 77.6144
SP 14	12.9750, 77.6072	**CP 14**	12.9853, 77.6241
SP 15	12.9763, 77.6148	**CP 15**	12.9960, 77.6144
SP 16	12.9788, 77.6191	**Terminal**	12.9702, 77.5853
SP 17	13.0037, 77.6187		

5.2.4. Time window verification

Make sure the routes chosen adhere to the time windows $[\psi_x, \zeta_x]$. If constraints are violated, adjust paths iteratively to find feasible alternatives.

5.3. Regression assessment using decision trees in real-time

5.3.1. Prototypical training

Use past trip time data for training a Decision Tree Regression model (Tm), energy usage (W_{xy}), traffic and road conditions.

5.3.2. Data emulating and excruciating

Incorporate important aspects like traffic on the roads and the availability of charging outlets (co_y), and swiftness (v) help enhance model forecasts.

5.3.3. Forecast of route feasibility

Utilize the learned model to forecast energy usage (W_{xy}) and travel time (Tm) for possible routes, allowing for a continuous assessment of their viability.

5.3.4. Dynamic modification

Real-time route adjustments based on forecasts are made to guarantee adherence to limitations such as battery capacity (w_x) and delivery time windows $[\psi_x, \zeta_x]$.

5.4. Unified method for augmenting paths

5.4.1. Preliminary Path choice

To find potential routes that meet basic restrictions like battery life (w_x), apply Dijkstra's Algorithm and time windows $[\psi_x, \zeta_x]$.

5.4.2. Refinement through forecasts

Refine and improve initial routes using Decision Tree outputs to save energy and trip time (Tm).

5.4.3. Iterative optimization

Combine the results of Dijkstra's Algorithm and Decision Tree Regression to continuously improve pathways and find the optimum tradeoffs between time (Tm), energy (W_{xy}), and costs.

5.4.4. Data modification and enhancement

5.4.4.1. Model updates. Retrain the Decision Tree model regularly using updated data to account for changes in the charging infrastructure or traffic patterns.

5.4.4.2. Complex algorithms. To improve prediction accuracy, additional machine learning methods should be explored.

6. Simulation results and analysis

Electric automobiles are currently transforming urban transportation in developing countries. According to this analysis, India is one of the emerging nations capable of effectively addressing challenges in its major cities. India's transportation infrastructure is evolving due to a rising trend toward EAs, driven by government incentives and environmental concerns. Despite the country's severe routing issues, the increasing adoption of EAs offers opportunities to improve travel routes and charging networks, ultimately leading to more efficient urban transportation and lower carbon emissions. The government is also deploying battery swapping for public EA transportation to enable quick and efficient vehicle turnover. This method minimizes downtime and ensures continuous operation for electric buses and other public vehicles by exchanging depleted batteries for fully charged ones. However, the public has yet to widely adopt this approach for four-wheeled automobiles. Not all public transit systems utilize the EAPPTF technique. This study aims to optimize EA route planning by considering time frames, battery capacities, and traffic conditions.

By effectively optimizing EA route planning based on traffic flow, charging station availability, and variable travel times, this approach seeks to enhance the overall sustainability and efficiency of urban transportation. Bengaluru City

Table 3. Main charging points and client spots in Hyderabad city.

Charging outlet points	Charging outlet locations	Customer points	Customer locations
SP 1	17.4458, 78.4775	**CP 1**	17.4519, 78.4661
SP 2	17.4384, 78.4537	**CP 2**	17.4535, 78.4463
SP 3	17.4284, 78.4322	**CP 3**	17.4344, 78.4203
SP 4	17.4345, 78.4582	**CP 4**	17.4263, 78.4609
SP 5	17.4170, 78.4508	**CP 5**	17.4091, 78.4425
SP 6	17.4085, 78.4552	**CP 6**	17.4132, 78.4633
SP 7	17.4120, 78.5060	**CP 7**	17.4273, 78.5173
SP 8	17.4042, 78.5079	**CP 8**	17.4000, 78.5153
SP 9	17.3923, 78.5179	**CP 9**	17.3813, 78.5293
SP 10	17.4028, 78.5417	**CP 10**	17.3896, 78.5480
SP 11	17.3569, 78.5479	**CP 11**	17.3622, 78.5300
SP 12	17.3693, 78.5584	**CP 12**	17.3765, 78.5726
SP 13	17.4157, 78.5606	**CP 13**	17.4258, 78.5641
SP 14	17.4410, 78.5557	**CP 14**	17.4512, 78.5488
SP 15	17.4456, 78.5612	**CP 15**	17.4562, 78.5739
SP 16	17.4733, 78.5656	**CP 16**	17.4804, 78.5979
SP 17	17.4672, 78.5618	**CP 17**	17.4791, 78.5446
SP 18	17.5036, 78.5384	**CP 18**	17.4903, 78.5232
		Terminal	17.4278, 78.4891

(BC) serves as a solid foundation for this work's real-time EA route optimization, which aims to reduce congestion and improve delivery efficiency through its support for a rapidly growing urban population and complex traffic patterns. BC has made significant progress in adopting EAs due to various programs promoting them as a sustainable way to lower air pollution and dependence on fossil fuels. The city is an ideal location to examine EA routing challenges because its infrastructure, including charging stations and EV-friendly policies, continues to support electric mobility. Implementing and improving EAPPTF-based solutions can be challenging due to the network's complexity, frequent traffic congestion, and changing road conditions, but they are feasible. The real-time traffic data provided by city monitoring systems provide the dynamic information necessary for accurate route optimization. This is especially important for EVs, as they must account for battery life and time window restrictions. Thanks to its commitment to sustainability and advanced digital infrastructure, BC provides a valuable real-world model for urban logistics and transportation research related to the proposed EAPPTF challenge. Table 2 lists the depots (DPT), major charging stations, and customer sites in BC that require EA services.

Hyderabad City (HC) faces complex transportation demands due to its role as an economic and technological hub in India as well as its rapid urban growth. With a growing population and increasing demand for transportation, especially in the retail and e-commerce sectors, the need for efficient routing solutions that reduce trip times, energy consumption, and operational costs is evident. Consequently, HC is an ideal place to evaluate the proposed EAPPTF solutions. The city's road network includes busy arterial highways, narrow lanes in historic neighborhoods, and modern expressways. Route optimization is complicated by infrastructure diversity, fluctuating traffic, and ongoing construction projects. This is particularly true when considering time constraints, real-time traffic data, and EA battery limitations.

These features make HC an excellent location to test and deploy algorithms capable of real-time adaptation to this complexity, such as the Dijkstra algorithm. The city is increasingly adopting EAs to promote eco-friendly mobility and reduce pollution. The expansion of EA infrastructure, including public and private charging stations, creates an ideal environment for integrating real-time data and addressing urban EA challenges. Accessible infrastructure allows testing of solutions that monitor battery levels and charging station availability in real-time. Moreover, HC's commitment to smart city initiatives, such as data analytics and live traffic monitoring, provides the data and insights needed to optimize EA routes. This ensures that EAPPTF solutions remain adaptable and responsive to the constantly changing urban landscape. Table 3 lists the main charging stations and key customer sites in HC.

Table 4 presents a comparison of the key differences between Hyderabad and Bengaluru's Electric Automobile Path Planning with a Time Frame (EAPPTF) in terms of city features, data collection and analysis, and outputs.

6.1. Approaches for solving the electric automobile route planning problem

6.1.1. Way planning conundrum in Bengaluru city

Path planning in BC involves determining the optimal routes for vehicles, especially EAs, while accounting for complex road networks, real-time traffic, and changing conditions such as traffic jams and road closures. To meet delivery windows, it is crucial to reduce travel time and energy consumption due to dense urban infrastructure and fluctuating traffic patterns. Overcoming this challenge requires developing routes that strike a balance between sustainability, efficiency, and battery management by integrating real-time data with sophisticated algorithms. Addressing these elements is essential for solving the proposed EAPPTF problem, and finding solutions for these instances is likewise critical.

6.1.1.1. Instance with a single electric automobile. The most efficient route for BC with a single EA that visits multiple charging stations and clients is shown in the graph in Figure 4. The EA follows a designated path that minimizes travel distance and optimizes battery performance, starting from the depot. The strategic placement of charging stations along

Table 4. Comparison of EAPPTF performance between Bengaluru and Hyderabad.

Category	Bengaluru	Hyderabad
City characteristics		
Traffic patterns	High traffic congestion occurs, especially during peak hours.	Congestion is moderate to high, but slightly better than in Bengaluru.
Population density	Higher density in central areas results in complex routing.	Lower density in suburban areas, but density is growing in central zones.
EA adoption rates	EA adoption is relatively high due to government policies and subsidies.	EA adoption is increasing, though slightly behind Bengaluru..
Measures of collecting information		
Delivery demand	Traffic levels are high due to the city's strong IT and e-commerce sectors.	Demand is moderate but rising, particularly in suburban areas and tech hubs.
Charging infrastructure	Relatively well-developed but unevenly distributed across the city.	There are gaps in the emerging infrastructure in some suburban and outlying areas.
Traffic and route data	Complicated by ongoing roadwork projects and traffic jams.	Less congested and easier to predict than in Bengaluru.
Components of evaluation		
Routing optimization	Aimed at reducing delays caused by regular traffic jams.	Focused on traveling farther distances while experiencing mild traffic disruptions.
Charging schedules	Shorter travel distances and significant traffic delays may require frequent recharging.	Improved range management with fewer recharging interruptions.
Fleet composition	Densely populated locations are best suited for compact EAs.	Demand from the suburbs has resulted in a mix of EA sizes.
Yields		
Delivery efficiency	Traffic and time windows limit high efficiency.	Somewhat improved efficiency is achieved due to wider windows and traffic control.
Cost reduction	There are significant reductions in emissions and fuel use, but this may result in increased time costs.	Cost and time savings are effectively balanced.
Emission reduction	A substantial decrease is observed as a result of heavy reliance on EA fleets.	Depending on adoption rates, the reduction may range from moderate to significant.

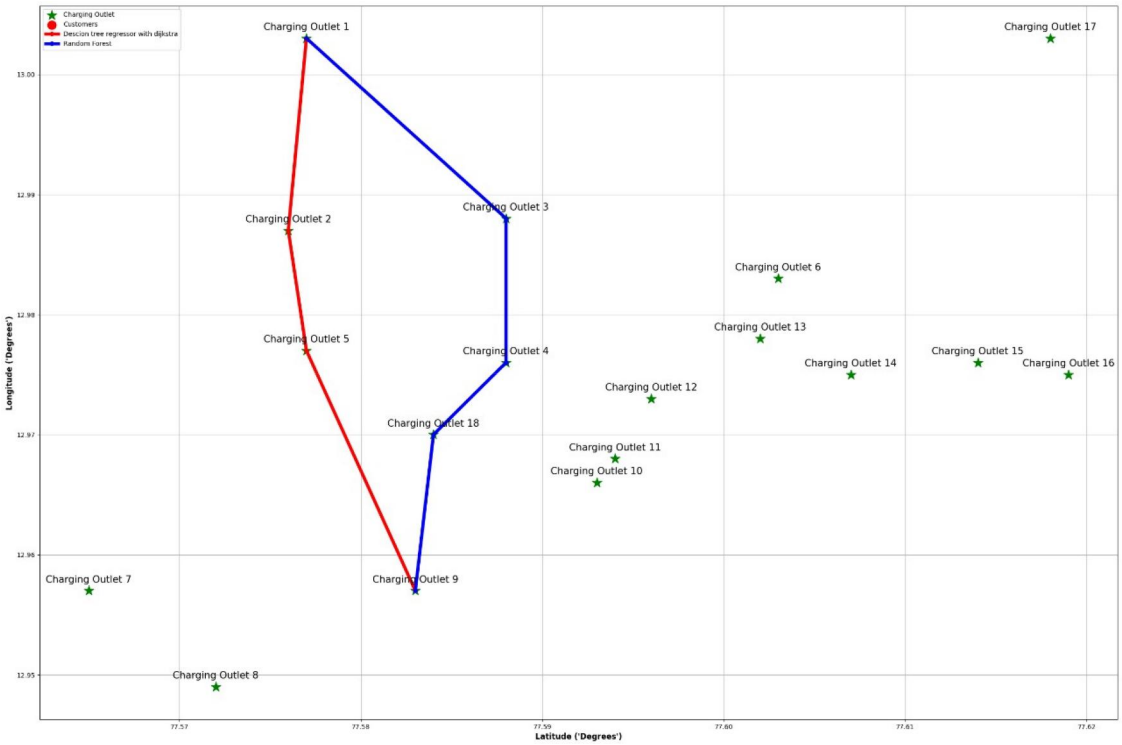


Figure 4. Best route for a single electric automobile with major charging outlets in Bengaluru, India.

the route allows the EA to recharge easily, enabling it to continue fulfilling its delivery and service commitments.

This carefully planned route demonstrates how EAs can provide cost-effective and environmentally friendly transportation. EAs contribute to a greener future by reducing energy consumption and dependence on fossil fuels. To further improve EA performance, considerations such as battery capacity, charging schedules, traffic patterns, and customer preferences must be taken into account. These additional constraints are necessary for designing real-time solutions, although they also introduce increased complexity.

6.1.1.2. Case with multi-electric automobiles. An optimal route plan for three EAs to efficiently serve multiple clients

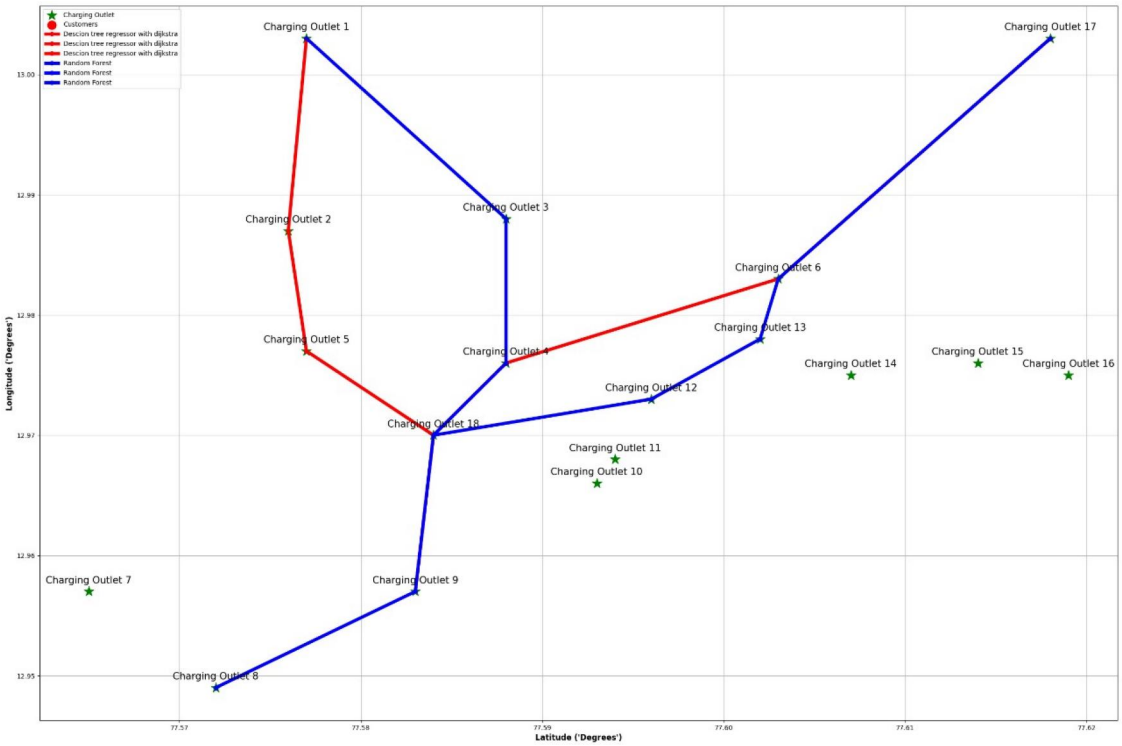


Figure 5. Best route for three electric automobiles in Bengaluru, India, with major charging outlets.

while minimizing energy use and travel time is shown in Figure 5. Beginning at a central depot, each EA follows a different route carefully designed to include stops at various recharge stations and client locations. The best routes are likely identified using algorithms such as Dijkstra and/or decision trees, which account for factors like distance, traffic, and battery life. By planning routes meticulously, optimizing resource use, and lowering overall operational costs, this approach aims to balance workload distribution among the three EAs.

The performance of the EA fleet depends heavily on the strategic placement of charging stations along the routes. These stations enable the EAs to recharge, ensuring continuous and uninterrupted delivery. The solution enhances overall fleet efficiency and reduces downtime through an optimized charging schedule.

6.1.1.3. Case with single electric automobile and customers (CSEA with customers). To efficiently deliver goods to various clients while reducing energy consumption and travel time, the graph in Figure 6 illustrates an optimal route for a single EV. This route, starting from a central depot, includes charging stations at key points to ensure sufficient battery capacity for completing all deliveries. Algorithms like Dijkstra and decision trees, which evaluate factors such as distance, traffic, and battery capacity, are likely employed to select the best route. This planning underscores how EVs can provide effective and sustainable services while reducing total trip distance and energy consumption.

To ensure that the EA reaches all assigned clients with enough battery charge, the Dijkstra algorithm seeks to minimize the total travel distance. Route planning also considers real-time charging station availability allowing continuous adjustment of routes. This approach maximizes the vehicle's energy efficiency, route effectiveness, and optimizes the placement of charging outlets in BC.

6.1.1.4. Scenario of the optimal path for a single electric vehicle visiting charging outlets and customer locations under constraints. An optimal path for a single EA to visit multiple clients while considering various restrictions is shown in the graph of Figure 7. The route ensures that all deliveries are completed efficiently and effectively. It guarantees on-time delivery, reduces travel distances, and conserves energy. To maintain sufficient battery capacity, the EA recharges at designated stations. Routes can also be adjusted in real time to avoid traffic congestion and minimize delays. This strategy highlights how EAs can operate reliably and efficiently despite multiple constraints. Factors like traffic conditions, delivery schedules, limited battery capacity, and charging station availability play a significant role in route planning. By addressing these considerations, EA usage is maximized, and operational costs can be reduced.

6.1.2. Path planning problem of Hyderabad city

Because of specific constraints such as limited battery capacity and dependence on charging infrastructure, planning for EAs in Hyderabad city requires special attention. By considering these limitations, routes become more practical and efficient, ensuring that service or delivery commitments are met without exceeding operational limits.

6.1.2.1. Case with a single electric automobile in Hyderabad city. An effective route for a single EA to visit multiple

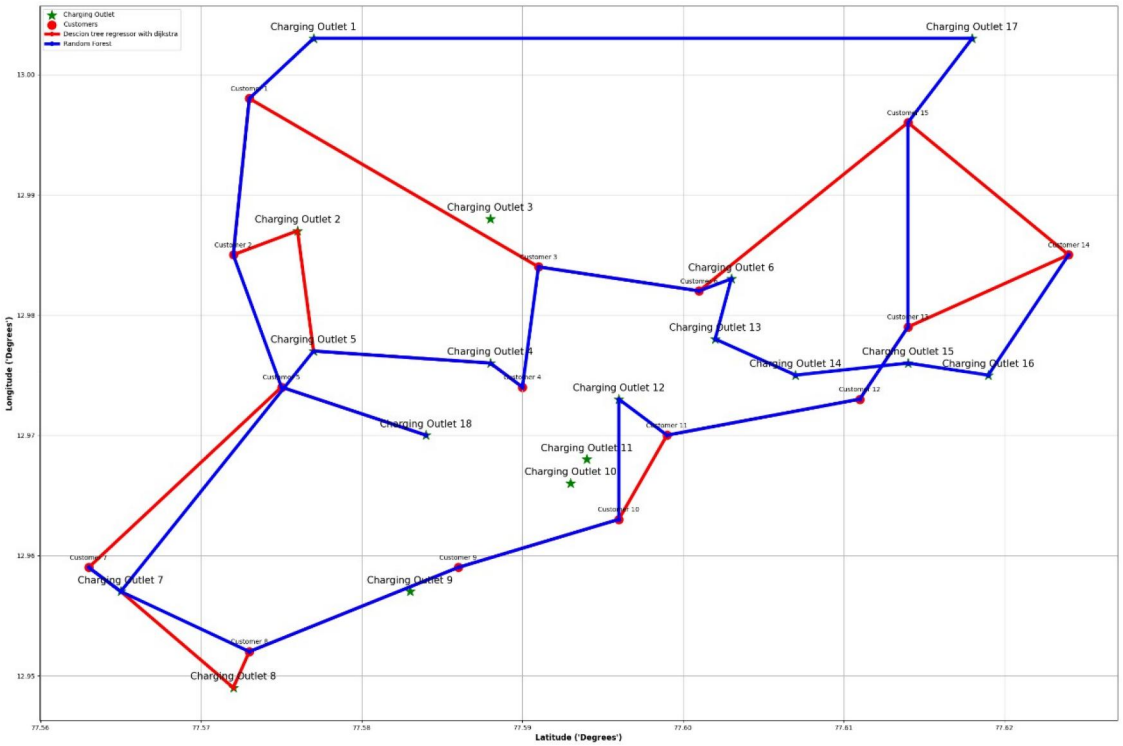


Figure 6. The best route for an EA through Bengaluru's main customer locations and charging outlets.

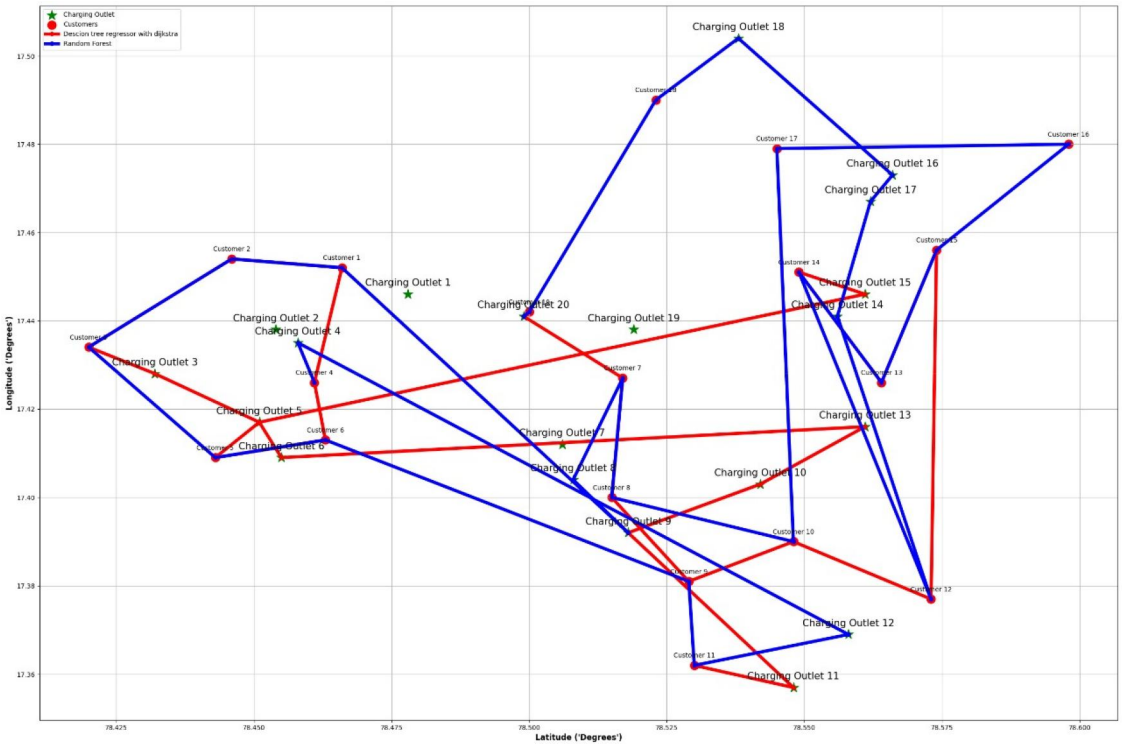


Figure 7. The best route for electric automobiles includes major charge points and customer terminals within Bengaluru City Limits.

charging stations is shown in Figure 8. The EA starts from the depot and follows a carefully planned route that aims to at minimizing travel distance and maximizing battery efficiency. Along the way, charging stations are strategically located for easy recharging, enabling the EA's smooth journey. This approach highlights the importance of route optimization in promoting environmentally friendly transportation. EAs contribute to environmental goals by reducing energy consumption and dependence on fossil fuels. Incorporating these constraints and refining the strategy further enhances operational efficiency.

6.1.2.2. Case with multiple electric automobiles. An optimal route plan for several EAs) to visit various charging stations

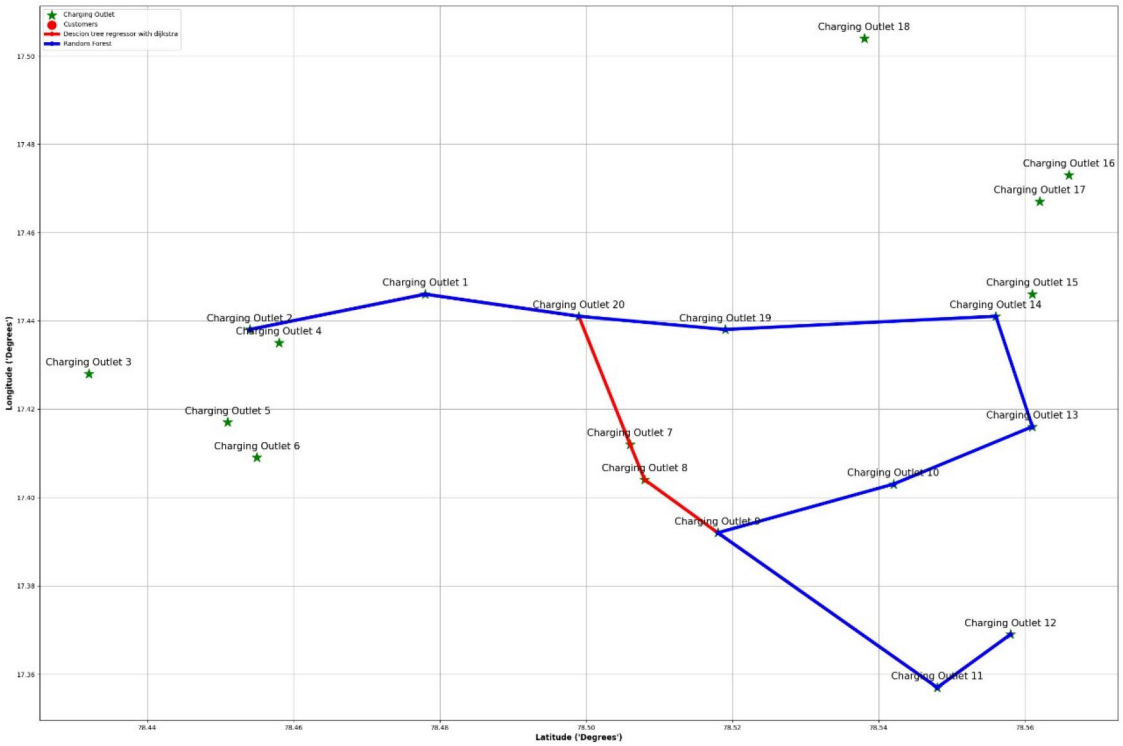


Figure 8. The best route for a single electric automobile featuring multiple recharge outlets in Hyderabad.

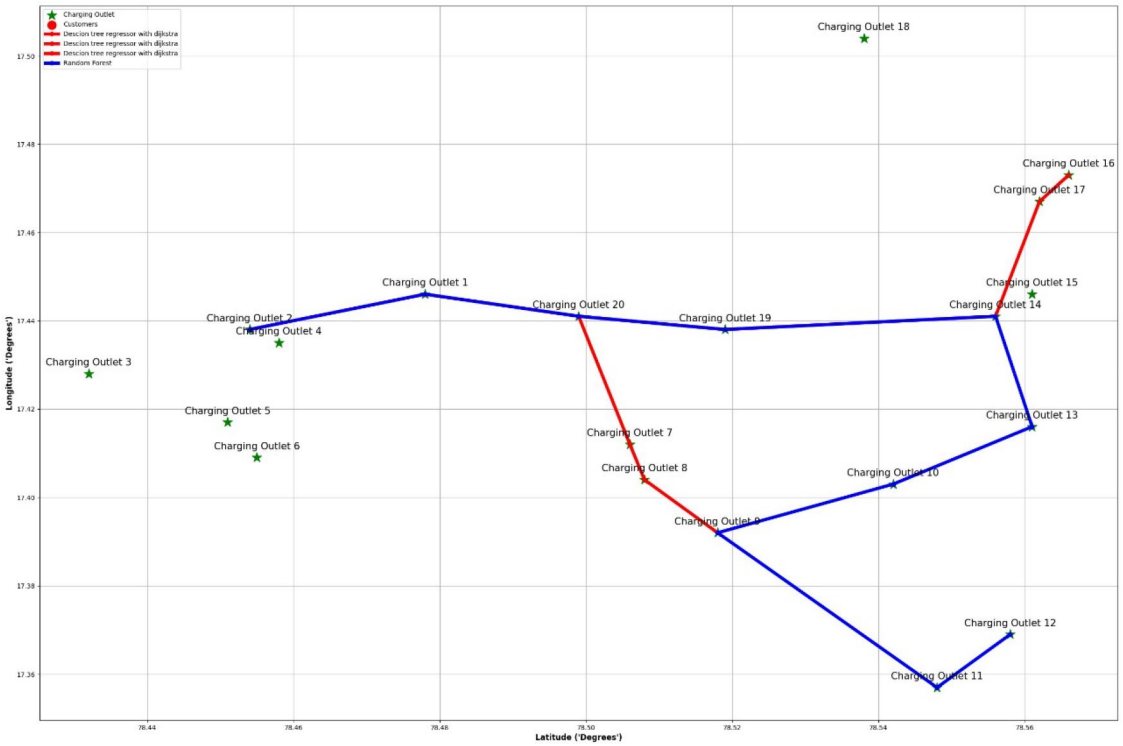


Figure 9. An optimal route for three electric automobiles with multiple recharge outlets in Hyderabad.

efficiently is illustrated in Figure 9. Each EA starts from a central depot and follows a designated route designed to minimize travel distance and ensure efficient battery usage. Algorithms like Dijkstra and decision trees, which consider factors such as distance, traffic, and battery capacity, are likely used to identify these optimal routes. Through detailed route planning, the approach promotes effective resource utilization, and reduces total operating costs, the system aims to distribute tasks among the EAs. Proper placement of charging stations along these routes is essential for the fleet's efficient operation. EAs can recharge at these stations and continue their trips without interruption. The proposed method maximizes fleet efficiency and reduces downtime by optimizing charging schedules.

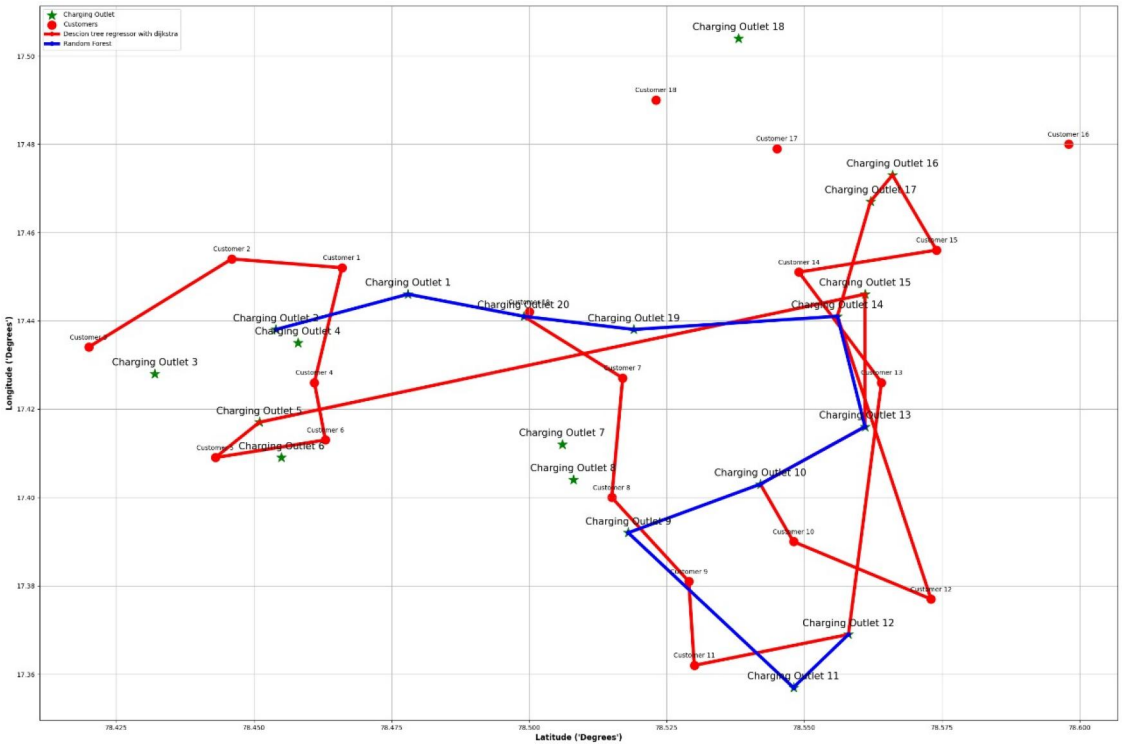


Figure 10. An optimal route for electric automobile with Hyderabad's main charging outlets and customer locations.

6.1.2.3. Case with single electric automobile. An optimal route plan for a single EA to visit multiple clients and charging stations is illustrated in Figure 10. The EA starts from the depot and follows a predefined route to minimize trip distance and optimize battery usage. With charging stations positioned along the route, the EA recharges at optimal times and completes its tasks efficiently.

This carefully planned route enhances the sustainability and efficiency of last-mile delivery services. EAs support a greener, more sustainable future by reducing energy consumption and reliance on fossil fuels. The Dijkstra algorithm helps minimize the total journey distance, ensuring the EA reaches all assigned clients with sufficient battery charge. The route planning system also considers factors such as charging station availability to improve the delivery process and adapt to real-time changes.

6.1.2.4. Case with a single electric automobile with charging outlets and customer points. An optimal path for a single EA to efficiently serve multiple clients while considering various limitations is illustrated in Figure 11. Route planning is significantly influenced by these constraints, including limited battery capacity, delivery time windows, charging station availability, and traffic conditions. The optimal route ensures that the EA minimizes energy consumption and prolongs battery life while completing all deliveries efficiently.

The route is designed to meet all delivery deadlines, reduce energy consumption, and minimize travel distance. To maintain sufficient battery power for deliveries, the EA recharges at designated charging stations. Real-time route adjustments are also possible for minimizing delays and avoid traffic congestion. This optimal route planning demonstrates that, despite several challenges, EAs can deliver goods efficiently and on time. Considering these factors helps lower operating costs and enhances the utilization of EAs. The software automatically adjusts for traffic conditions and charging station closure if a station malfunctions. Any vehicle issue that makes the vehicle unusable will also be updated in the system to maintain accurate fleet operations.

6.2. Comparative analysis of total travel distances and duration in various considered cities

The most effective times for covering the shortest distances in Hyderabad and Bengaluru are shown in Tables 5 and 6, which evaluate EA routes in EAPPTF. This case study examines Indian cities using real-time data. According to Table 5, an electric automobile (EA) takes the quickest and least expensive route to reach one of these seventeen charging outlets (COs). There are fifteen consumers (Cs) that each EA visits; if needed, it travels the optimal distance and time to a nearby CO. This scenario for Hyderabad and Bengaluru performs effectively in real-world situations using the proposed machine learning technique.

The suggested hybrid decision tree and Dijkstra method, combined with random forest ML techniques for the EAPPTF problem, are simulated across various scenarios, as shown in Figures 4–11. For different vehicle counts, the hybrid approach efficiently finds the best routes, nearest charging outlets, and consumer locations in both cities. According to Tables 5 and 6, the decision tree and Dijkstra algorithms outperform the random forest algorithm by delivering the most effective distances and durations for the EAPPTF problem. These algorithms can automatically adapt

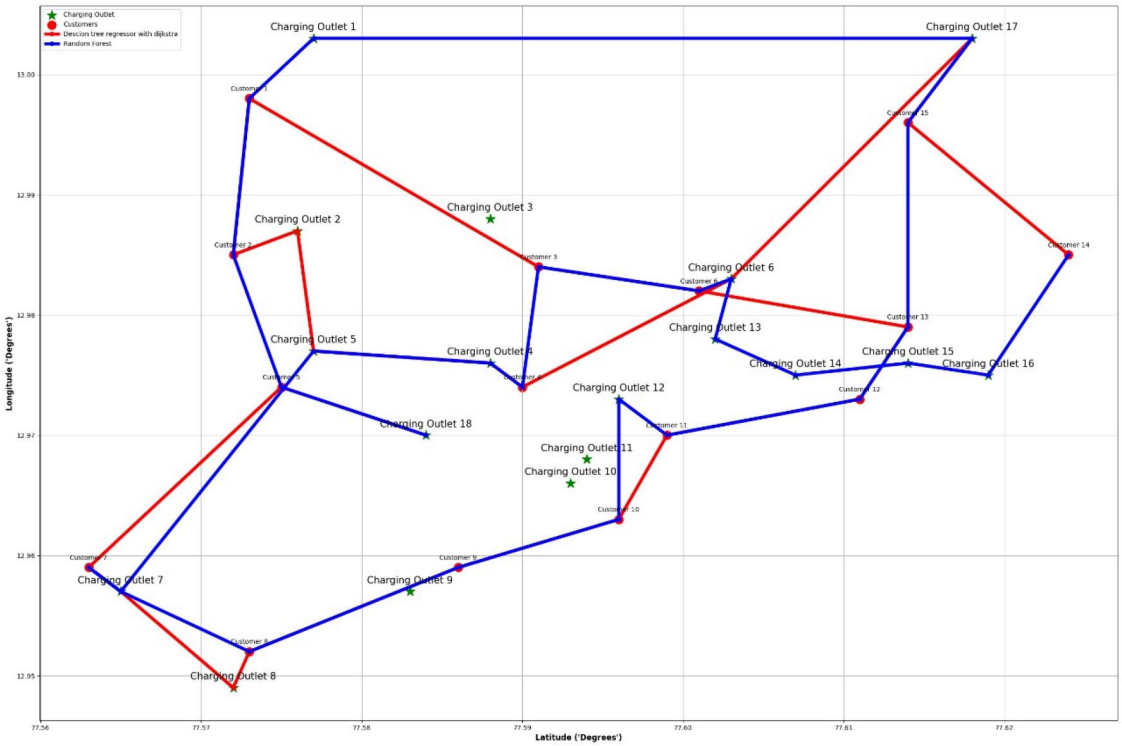


Figure 11. An optimal route for electric automobiles featuring major charging stations and consumer locations, considering Hyderabad's operational constraints.

Table 5. The optimal distance and time achieved using decision tree and Dijkstra algorithms compared with random forest in Bengaluru.

	Decision tree & Dijkstra Algorithm		Random Forest Algorithm	
EAPPTF parameters	Distance (km)	Duration (h)	Distance (km)	Duration (h)
CO 2–CO 16	37.90	01.32	38.99	01.35
EA 5–CO 17	43.50	07.43	45.72	07.61
EA 1 – CO 17 – C 15	110.07	04.62	111.62	04.74

Table 6. The optimal distance and time achieved using decision tree and Dijkstra algorithms compared with random forest in Hyderabad.

	Decision tree & Dijkstra Algorithm		Random Forest Algorithm	
EAPPTF parameters	Distance (km)	Duration (h)	Distance (km)	Duration (h)
EA 1 – CO 17	16.35	00.37	18.04	00.30
EA 5 – CO 17	22.20	11.80	23.96	12.75
EA1 – CO 17 – C15	139.93	05.26	142.05	05.28

to changing traffic densities and new traffic patterns by updating the training data as needed.

6.3. Proposed decision tree-based EAPPTF

An innovative method called Electrical Automobile Path Planning using Time Frame (EAPPTF) has been developed to optimize the navigation and charging needs of EA users in public transportation systems. Regression tree models enable accurate identification of charging outlets while reducing the time spent searching for them. In the selected cities, this approach ensures shorter wait times and greater user convenience by predicting the availability of charging outlets. Additionally, EAPPTF offers real-time routing updates, allowing users to access stations efficiently in a dynamic urban environment. This technique, which combines predictive modeling with time-frame-based path planning, is a vital tool for promoting sustainable and effective EA adoption in public transit networks. It enhances system reliability, minimizes delays, and improves overall satisfaction for EA users.

The graph in Figure 12 compares various EAPPTF-related parameters between Hyderabad and Bengaluru. It highlights differences in their delivery demands, EA adoption, and traffic patterns. Hyderabad excels in fleet composition and pollution reduction, while Bengaluru outperforms improves charging scheduling and traffic optimization. Both cities are nearly equal in delivery efficiency and cost reduction. The analysis emphasizes each city's unique advantages and potential development areas.

This work primarily focuses on implementing the EA four-wheeler sharing system in most emerging nations, with a comparison of the two cities. The EAPPTF approach has been successfully tested in Indian cities, offering other other developing nations the chance to adopt similar systems in busy urban environments.

7. Conclusion

This work presents an approach that effectively addresses the challenges of routing electric automobiles in urban environments. The Electric Automobile Path Planning with Time Frame (EAPPTF) tackles the electric automobile routing problem by combining graph-based algorithms and machine learning techniques. It greatly improves route planning, traffic management, energy savings on inclines, and

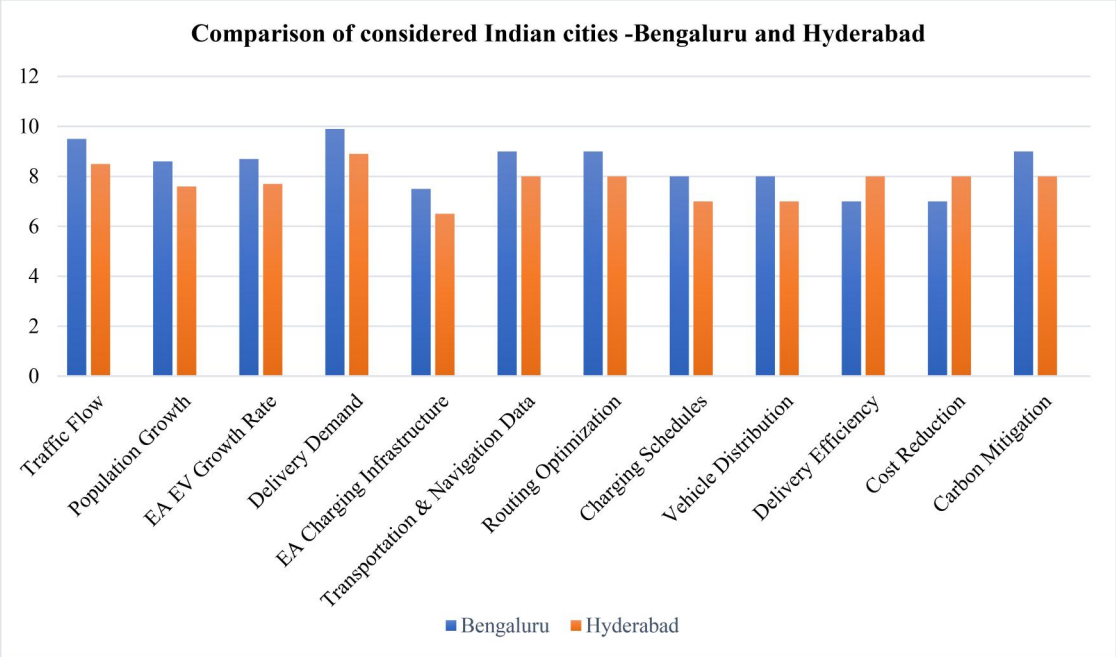


Figure 12. Graphical comparison of traffic patterns between Hyderabad and Bengaluru for electric automobile path planning.

delivery schedules that adjust in real-time, considering factors like battery life and charging stops. Dijkstra's algorithm, enhanced by Decision Tree Regression, finds the best paths connecting key cities in India. Real-time data from Indian cities such as Bengaluru and Hyderabad showcases the framework's flexibility. This results in more efficient routes, shorter travel times, and lower energy use, supporting eco-friendly urban mobility. Additionally, this method makes it easier to locate key customer sites. In public EA transit systems, it also helps in identifying important customer locations and nearby charging stations. By locating the most suitable charging stations nearby, this route optimization can be extended to four-wheeled electric vehicle sharing. The system estimates travel times and charging station availability within specific time frames by predicting traffic density. It constantly adjusts to changing city conditions, improving prediction accuracy and allowing precise estimates of trip duration and energy consumption. According to the results, the Dijkstra and decision tree algorithms outperform the random forest method, providing the best durations and distances for the EAPPTF problem. Future improvements might include integrating autonomous vehicles to optimize combined mobility options and using advanced machine learning techniques for enhanced prediction accuracy. This research paves the way for a sustainable future by highlighting the importance of data-driven strategies in developing cleaner, greener, and more efficient urban transportation systems.

Acknowledgement

The authors would like to thank the management of Chaitanya Bharathi Institute of Technology, Hyderabad, India, for providing ample space for conducting this research.

Author contributions

Vijay Raviprabhakaran wrote the manuscript text, conceptualization, methodology, validation, formal analysis, investigation, review, editing, visualization, and project administration. **Jesica Sarakonda, Vinay Udugula, and Nishitha Varma Datla** performed the data collection, validation of the data, and the study's conception, and assisted with a few sections of the manuscript. **Vijay Raviprabhakaran** wrote the first draft of the manuscript, and all authors provided feedback on its previous versions. All authors read and approved the final manuscript.

Disclosure statement

The author states that none of the work described in this study appears to have been influenced by any known competing financial interests or personal ties.

Funding

The author declares that no funds, grants, or other support were received during the preparation of this manuscript.

Data availability statement

No data was used for the research described in the article.

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